Safety Classifications

Lasers produce dangerous and invisible electromagnetic radiation. Radiation emissions during operation poses the risk of burns to the user's eyes and skin. Scattered and reflected radiation is equally as hazardous.

Since red radiation is refracted in the eye and concentrates on the retina, it is the most likely to cause eye damage. This high intensity radiation may burn the soft tissue of the retina, resulting in decreased vision or blindness. Safety glasses or protective eye covers must always be worn while operating or servicing the machine.

Laser-safe, optical density (OD) rated eyewear protects from direct, reflected, and scattered radiation to an extent; the intense laser radiation produced by the machine may still be harmful to a user's eyes with damaged or flawed personal protective equipment (PPE), so it is always recommended to check safety equipment thoroughly before use. Ensure the safety equipment has the correct OD rating for the machine, as safety glasses found on CO2 lasers are not suitable for the radiation emitted in equivalent YAG or solid-state lasers.

Although skin is more resistant to electromagnetic radiation than the eye, any affected areas can still be at risk for burns, with severity being dependent on exposure time and the degree of radiation.

Safety Precautions

Lasers produce an extremely fine, dust-like particulate when laser marking painted or anodized surfaces, or laser engraving various metals. It is recommended to keep the lens and enclosure dust and particulate free for optimal marking quality. Use ethyl or isopropyl alcohol on a clean rag to regularly wipe the surface of the lens.

Proper ventilation and use of the fume extractor minimizes particulate and gas inhalation risks for the operator. We recommend replacing the filter elements per the time intervals specified in the <HLS0E user manual>. Additional filter elements can be purchased at HaasTooling.com.

For any abnormal scenarios or failures, immediately press the E-stop button to disable the machine.

⚠ **Only a certified technician is allowed to repair this machine.**

⚠ **All information provided for reference and illustrative purposes only. Haas Automation cannot be held responsible for repairs you perform. Only services and repairs provided by authorized Haas Factory Outlet technicians are guaranteed.**

⚠ **Only an authorized Haas Factory Outlet distributor should service or repair a Haas machine that is protected by the original factory warranty. Servicing by any other party voids the warranty.**

Fire precautions should always be observed when using laser machines. Since fiber laser machines have high peak output powers, it is recommended to remove or displace any nearby flammable materials or fluids as they pose considerable fire risks during operation. An easily accessible fire extinguisher is required in the room of operation.

⚠ **IMPORTANT:** When using the side access windows to facilitate larger parts, the openings must be adequately covered to absorb all reflected light and protect operator eyesight. The windows are not equipped with a safety sensor. Lasers produce dangerous and invisible electromagnetic radiation. Radiation emissions during operation poses the risk of burns to the user's eyes and skin. Scattered and reflected radiation is equally as hazardous.

Safety Decals

The Haas factory puts decals on your machine to quickly communicate possible hazards. If decals become damaged or worn, or if you need additional decals to emphasize a particular safety point, contact your Haas Factory Outlet (HFO).

NOTE: Never alter or remove any safety decal or symbol.



Be sure to familiarize yourself with the symbols on the safety decals. The symbols are designed to quickly tell you the type of information they give:

- Yellow Triangle - Describes a hazard.
- Red Circle with Slash-Through - Describes a prohibited action.
- Green Circle - Describes a recommended action.
- Black Circle - Gives information about machine or accessory operation.

Example Safety Decal Symbols: (1) Hazard Description, (2) Prohibited Action, (3) Recommended Action.

You may find other decals on your machine, depending on the model and options installed. Be sure to read and understand these decals.



You can find the majority of the hazardous, prohibited action, and recommended action symbols in this particular decal.

Hazard Symbols - Yellow Triangles



There is a fiber laser equipped in the machine that emits radiation and causes burns if you have direct contact with the laser when in operation.

Take care by avoiding contact with the laser by keeping all windows closed and locked and door closed. Wear the appropriate safety gear when operating the laser marker.



There are high voltage components on the machine that can cause electrical shock.

Take care to avoid opening electrical enclosures unless components are de-energized or proper personal protective equipment is worn. Arc flash ratings are on the nameplate.

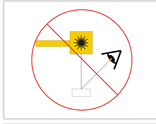


Machining operations can create hazardous dust. This is a function of the materials being lasered and the laser's power/speed settings.

It is up to the owner/operator of the machine to determine if personal protective equipment such as safety goggles or a respirator is required.

Lasers are equipped with the fume extractor system to prevent the inhaling dust.

Prohibited Action Symbols - Red Circles with Slash-Through



Do not look at the laser without the use of OD safety glasses or OD safety goggles.



Keep your hands and body away from the laser when in operation.

Recommended Action Symbols - Green Circles



We recommend wearing OD safety glasses or OD goggles when you are near a machine. The laser marker is equipped with Raised Safety Windows. Specifications are listed below.

Optical Density (OD) - Rated Laser Safety Windows

>930-1030nm DI ABS

>1030-1090nm DI ABS + IR ABTY

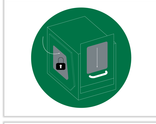
>1090-1130nm DI ABS

930-1030nm OD>5

1030-1090nm OD>7

1090-1130nm OD>6

1130-1700nm OD>5

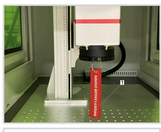


Keep the machine door closed and the windows closed and locked.



Always have the fume extractor on when operating the laser to avoid inhaling particles.

Other Safety Decals



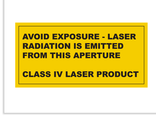
Lens Decal

The laser is equipped with a lens cover to avoid any damage that can occur to the lens. Before operating the laser, you must remove the lens cover. This decal tag is attached to lens cover and is there to serve as a reminder to remove the lens cover before using the laser.



Danger Decal

This decal shows the the dangers of the class IV laser that exposes you to laser radiation without the use of the appropriate personal protective equipment (PPE). This also states the max output and wavelength of the laser that is being worked with.



Laser Radiation Exposure Decal

This decal is shown to locate the laser head.

HLS0E Safety Features



Door Switch

The laser is equipped with a door switch [1] that keeps the operator safe by stopping the laser from operating if the door is open. The program will display a message when the door switch [1] is triggered and that will prevent the laser from operating.

⚠ **IMPORTANT:** The door should remained closed whenever the laser is in operation to prevent any accidents.



Emergency Stop

With the laser's E-stop, the user can push whenever they see an odd activity or possible danger when using the laser marker.



Safety Glass Door and Windows

The laser marker has CE rated glass windows and door.

⚠ **IMPORTANT:** Laser Marker should never be operated with the door and/or windows open.